Ch3: Evaluation Methodology :-

1. → challenges of evaluating foundation Models
2. → Understanding language modelling metrics
3. → Exact Evaluation
4. → AI as a Judge
5. → Ranking Models with Comparative Evaluation

Quality control AI outputs
Evaluation - to bring AI applications to reality.

- mitigate risks (system is likely to fail).
- uncover opportunities

FM's have LM's
To evaluate LMs → Cross-Entropy, perplexity.

→ FMs are open-ended, so Eval's are difficult to have.

AI as a Judge :
(Subjective evaluation) → the model, prompt that the AI judge uses

T① More intelligent AI models are difficult to evaluate.
Open-ended nature of FMs, no specific ground truth.
FMs are treated as black boxes.
Benchmarks get outdated very quickly
GLUE → Super GLUE
MMLU → MMLUPro.

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→ General-purpose models evaluation.
→ LLM evaluation papers have grown.
Anthropic funding for evals.

We need a Systematic approach to evaluation.

Understanding language Modeling Metrics:-

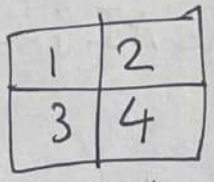
FMs have a LM
Metrics 4: Cross Entropy
Perplexity
BPC
BPB.

LM: Encodes statistical info.
The better it predicts the next token
→ The lower is the training cross entropy.

Entropy:- Measures how much information a token on average carries.
Higher entropy ↑
More bits are needed to represent a token.



1 bit



→ difficult to predict

2 bit

more info. (each token carries more info) but each token requires more bits to represent

↓
easier to predict for a LM as to what comes next

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Cross-Entropy:- Measures how difficult it is for a LM to predict what comes next in the dataset

A Model's Cross Entropy on train data
↳ ① Training data's predictability $H(P)$
② Divergence between the distribution captured by LM & the true dist. of train data $D_{KL}(P||Q)$.

$$H(P,Q) = H(P) + D_{KL}(P||Q)$$

Bits per character / Bits per Byte

unit CE/E: bits
say 6 bits is CE of LM: The LM needs 6 bits to represent each token.
But some models can use words/chars as tokens

$$BPC = \frac{6}{2} = 3$$

The problm of different character encoding schemes.
With ASCII each char: 7 bits
UTF-8 : 8 to 32 bits

So, BPB as more standardized.
↓ no. of bits a LM needs to represent one byte of orig train data. (each char is 7 bits)

$$BPB = 3/(7/8) = 3.43.$$

The LM can compress 8 bits of train text to 3.43 bits.

Perplexity

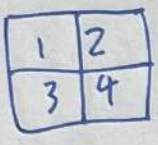
$$PPL(P) = 2^{H(P)}$$ P: True distribution of data

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$PPL(P, Q) = 2^{H(P,Q)}$

↓ learnt dist

Amt of uncertainty in pred. next token. / more uncertainty, more possible options for the next token.



PPL: $2^{2 \text{ bits}} = 4$

With units as nat, $PPL(P, Q) = e^{H(P,Q)}$.

Perplexity Interpretation & Use Cases:-

The more accurate a model can predict
→ The lower these metrics.

More structured data → lower perplexity
Bigger the Vocab → higher "
Longer the context length → lower ".

Say PPL = 3
If all tokens have equal chance of happening
⇒ Model has 1/3 chance of predicting next token correctly.

Guiding the training of language models
It's a good proxy for a model's capabilities

Post training can increase perplexity
Detecting data contamination
Used in deduplication of train data.
Can be used to detect abnormal texts.

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A model's perplexity w.r.t a text measures how difficult it is for the model to predict that text

$[x_1, \ldots, x_n]$ tokens

$$P(x_1, \ldots, x_n) = \left(\frac{1}{P(x_1, \ldots, x_n)}\right)^{\frac{1}{n}}$$

$$= \left(\prod_{i=1}^{n} \frac{1}{P(x_i | x_{i-1}, \ldots, x_1)}\right)^{1/n}.$$

3) Exact Evaluation :-

evaluate open-ended responses (text generations)

→ Functional Correctness : Code Generation.

Test Cases to check.

→ Pass @ K

10 P, solves 5, with K = 3

Pass @ 3 = 50%.

More code samples gen ↑ higher is the score

→ Similarity Measurements Against Reference data :

Evaluate AI's outputs against reference data.

Each ex: (input, reference responses (ground truths))

Requires ref. data

AI can also be used to generate ref. data.

Ways to measure the similarity between two open-ended texts :-

(1) Ask an evaluator to judge.

(2) Exact match.

(3) lexical similarity

(4) Semantic Similarity.

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So these are hand-designed metrics.

- Retrieval & search
- Rank (similarity wrt a query)
- clustering
- Anomaly detection
- Data deduplication

Exact Match :- Not well suited for complex tasks.

Lexical Similarity: How many tokens two texts have in common
- Edit distance.
- n-gram similarity

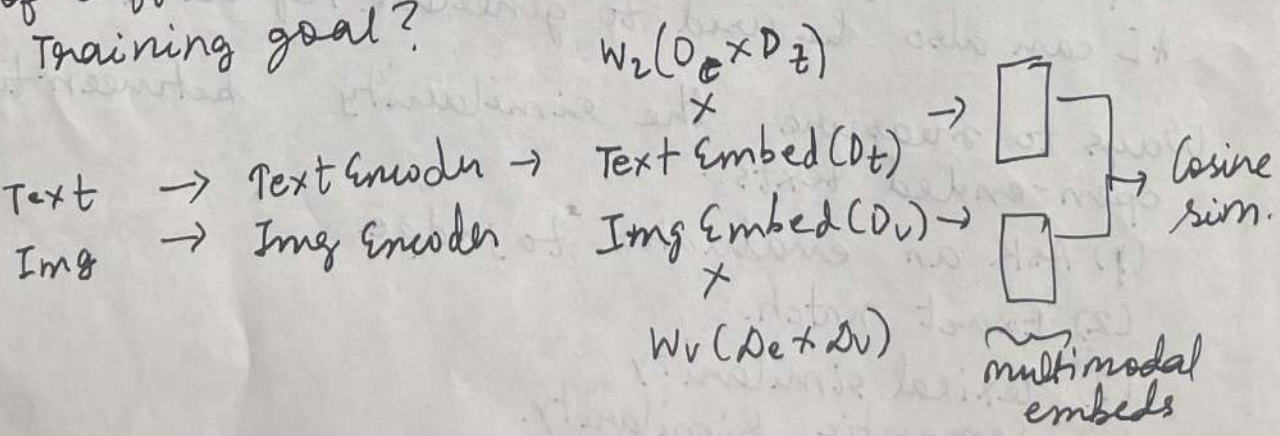
One drawback is curating a comprehensive set of reference responses.
- → Also low-quality reference data.
- → Doesn't always guarantee optimizing for functional correctness.

Semantic Similarity:- embedding similarity.
dot product, cosine similarity.

Introduction to Embedding:- Good if similar texts have closer embeddings.

New frontier: Creat joint embeddings for data of different modalities. (CLIP)

Training goal?



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AI as a Judge:- AI to evaluate an AI.
More used in production.

Why? – Fast, cheap
– No ref. data need.

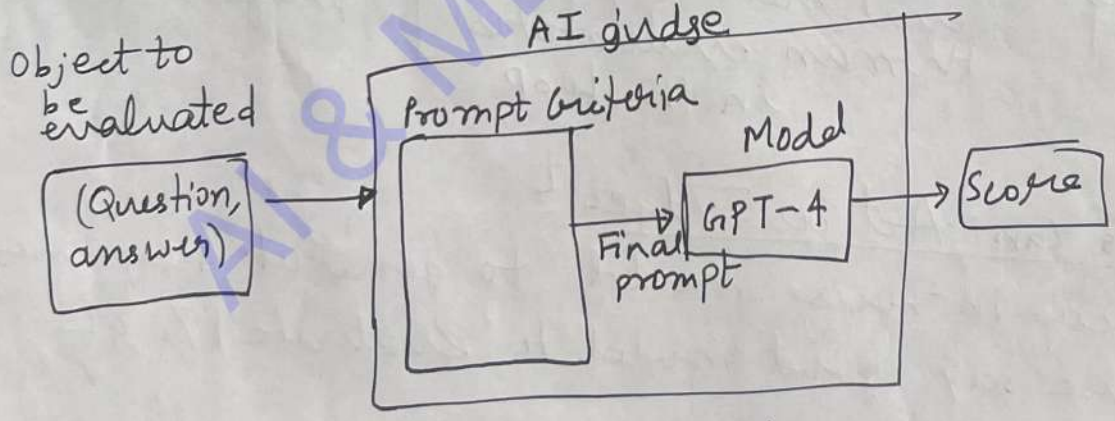
AI judges are strongly correlated to human evaluators.

How to use AI as a Judge?
→ Self-evaluation.
→ Compare gen & ref responses.
→ Compare two gen responses, and say which one is better.

AI as a judge metrics.

AI judge prompt – Task
– Criteria.
– Scoring systems.

prompts with examples perform better.



Limitations of AI as a Judge:-

Inconsistency:- AI is probabilistic.
Include evaluation examples in prompt
Twiddle sampling variables.
(But makes prompts longer)
more latency & Inference costs.

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Criteria ambiguity:- AI judge metrics aren't standardized.

AI judges are AI applications.

Increased costs & latency:- Using powerful models to evaluate responses can be expensive

We can use weaker models as AI as a Judge.
It can add ~~prod~~ latency.
Reduced risk though.

Biases of AI as a Judge:-
AI judges have self-bias
First-position bias. (repeat Qs with different orderings)
verbosity bias.
privacy issues

[AI judges + Exact eval methods/ Human evaluators in the loop] ✓ preferred.

What Models can Act as Judges?
→ Use a cheap in-house model to generate responses and use say good model to evaluate 1% of the responses

challenges: Who will be the judge for the strongest model?

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using a model to evaluate itself
↳ self-evaluation / self-critique.

Can Judge being weaker than the model being judged?

Small, specialized judges (~~for~~ more reliable).

Specialized judges

- **Reward Model**: (prompt, response) pair, and score the goodness of response.
- **Reference-based judge**: (candidate resp, ref. response) similarity computation.
- **Preference Model**: I/P: (prompt, res 1, res 2) O/P: which response is better.

Ranking Models With Comparative Evaluation

Comp O/Ps :-
↳ gen by two diff. models
↳ same Model but with different sampling variables.

An Evaluator picks the winner.

Understanding which q's to give to the users.

In A/B Testing, a user see's O/P from one candidate model at a time.

Match Win Rate } Pairwise Model comparisons.

LMSys chatbot Arena leaderboards

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Challenges of Comparative Evaluation :-

↳ **Scalability bottlenecks**:- Quadratic comparisons in # models

Transitivity (Ranking algorithms)
A > B > C
↖ ↗
for AI models? assumption true

diff model pairs evaluated by diff evaluators on diff. prompts.

Efficient matching algo: for sampling matches.

↳ **Lack of standardization and QC**
- ↳ Anyone can game the rankings.
- ↳ No real-world grounding.

From Comparative performance to absolute performance:-

B is better than A
What can be said?
Cost-benefit analysis.

Future of Comparative Evaluation:-

↳ Easier to compare two O/Ps, than to give each O/P a concrete score.

↳ Captures human preference.

Next Ch: How to incorporate evaluation metrics in our evaluation pipeline?
"Build reliable evaluation pipelines."

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